Module 1: Biology - The Study of Life - Key Points

Learning Objectives

By the end of this module, you should be able to:

1. Distinguish living systems from nonliving examples using multiple characteristics.
2. Separate hypotheses, predictions, observations, and conclusions in a scientific test.
3. Place biological examples at the correct scale of organization.
4. Identify variables and evidence in a simple investigation.
5. Revise a biological explanation when new evidence appears.

Topics

- What counts as life
- Science as a way of testing
- Organization from cells to ecosystems
- Evidence, variables, and measurement
- How biological explanations improve

Key Terms to Know

- **Cell** - Basic unit of life bounded by a membrane.
- **Homeostasis** - Regulation that keeps internal conditions within a workable range.
- **Metabolism** - All chemical reactions that move matter and energy through life.
- **Hypothesis** - Testable explanation for an observed pattern.
- **Variable** - Factor that can change in an investigation.
- **Evidence** - Observation or measurement used to support or challenge a claim.
- **Theory** - Broad explanation supported by many lines of evidence.
- **Evolution** - Change in heritable traits of populations across generations.

Core Contents

1. Life is recognized by shared properties: cells, energy use, response, reproduction, regulation, and evolution.
2. Scientific claims are tested with observations, comparisons, variables, and repeatable reasoning.
3. Biological systems are organized across scales from molecules to biosphere.
4. Data become useful when measurements are tied to a clear question and source of error.
5. Strong explanations connect evidence to a claim and stay open to revision.

Study Tips

1. Start with the module's central claim before memorizing vocabulary.
2. Use the module's companion lab (see the course schedule pairing) as the evidence check.
3. Explain one mechanism in words, then redraw it as a simple diagram.
4. Answer retrieval questions without notes, then revise with the terms list.

Connected Lab

- `lab-01_measurement-methods.md`

Generated Visuals

- Module 01: Biology - The Study of Life Evidence Map
- Module 01: Biology - The Study of Life Reasoning Sequence
- Module 01: Biology - The Study of Life Retrieval and Lab Check